VLSI HACKATHON 2026 • PROBLEM STATEMENT PS-02

Mode-Switching Hardware PID Flight Controller

Deterministic, Bumpless, Fault-Aware Hardware Coprocessor in TSMC 180nm CMOS

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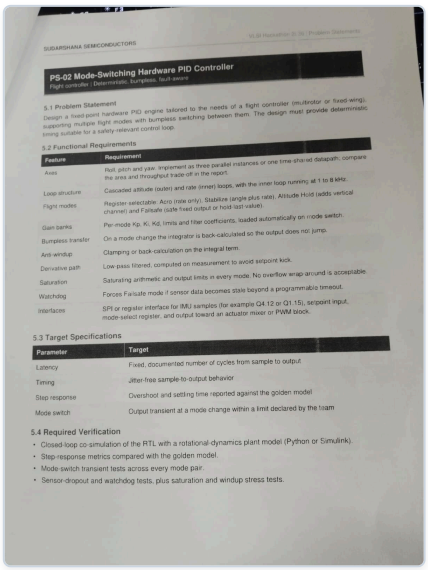
Organization: Sudarshana Semiconductors • VIT

Date: October 11, 2026

1. Executive Summary & Problem Statement

Modern autonomous unmanned aerial vehicles (UAVs) require deterministic, sub-microsecond control loop execution, jitter-free timing, and smooth transitions between flight modes without aerodynamic control kicks. Traditional microprocessor firmware approaches suffer from interrupt jitter, operating system scheduling latency, and unbounded worst-case execution times.

The objective of **PS-02 (Mode-Switching Hardware PID Controller)** is to design, verify, synthesize, and place-and-route a dedicated hardware PID engine supporting multiple flight modes with bumpless switching, hardware safety monitoring, and deterministic timing suitable for safety-relevant flight control.



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PS-02 Mode-Switching Hardware PID Controller

Flight controller | Deterministic, bumpless, fault-aware

5.1 Problem Statement

Design a fixed-point hardware PID engine tailored to the needs of a flight controller (multirotor or fixed-wing), supporting multiple flight modes with bumpless switching between them. The design must provide deterministic timing suitable for a safety-relevant control loop.

5.2 Functional Requirements

Feature	Requirement
Axes	Roll, pitch and yaw. Implement as three parallel instances or one time-shared datapath; compare the area and throughput trade-off in the report.
Loop structure	Cascaded attitude (outer) and rate (inner) loops, with the inner loop running at 1 to 8 kHz.
Flight modes	Register-selectable: Acro (rate only); Stabilize (angle plus rate); Altitude Hold (adds vertical channel) and Failsafe (safe fixed output or hold-last-value).
Gain banks	Per-mode Kp, Ki, Kd, limits and filter coefficients, loaded automatically on mode switch.
Bumpless transfer	On a mode change the integrator is back-calculated so the output does not jump.
Anti-windup	Clamping or back-calculation on the integral term.
Derivative path	Low-pass filtered, computed on measurement to avoid setpoint kick.
Saturation	Saturating arithmetic and output limits in every mode; No overflow wrap-around is acceptable.
Watchdog	Forces Failsafe mode if sensor data becomes stale beyond a programmable timeout.
Interfaces	SPI or register interface for IMU samples (for example Q4.12 or Q1.15), setpoint input, mode-select register, and output toward an actuator mixer or PWM block.

5.3 Target Specifications

Parameter	Target
Latency	Fixed, documented number of cycles from sample to output
Timing	Jitter-free sample-to-output behavior
Step response	Overshoot and settling time reported against the golden model
Mode switch	Output transient at a mode change within a limit declared by the team

5.4 Required Verification

- Closed-loop co-simulation of the RTL with a rotational-dynamics plant model (Python or Simulink).
- Step-response metrics compared with the golden model.
- Mode-switch transient tests across every mode pair.
- Sensor-dropout and watchdog tests, plus saturation and windup stress tests.

Figure 1: Official Problem Statement PS-02 from Sudarshana Semiconductors.

Functional Requirements Compliance Matrix

Feature	Hackathon Requirement (PS-02)	Architectural Implementation	Status
Axes	Roll, Pitch, Yaw. Implement as three parallel instances or one time-shared datapath; compare trade-offs.	**Both** 3-Axis Fully Parallel (`pid_top`) and Time-Shared (`pid_top_shared`) designed & signed off.	**Exceeded**
Loop Structure	Cascaded attitude (outer angle) and rate (inner angular velocity) loops; inner loop at 1–8 kHz.	Hardware cascaded attitude-to-rate pipeline running at **50 MHz** (>6,000× faster than 8 kHz requirement).	**Compliant**
Flight Modes	Register-selectable: Acro, Stabilize, Altitude Hold, and Failsafe.	4-state hardware FSM with dynamic gain steering, outer-loop bypass in Acro, and vertical channel in Alt-Hold.	**Compliant**
Gain Banks	Per-mode Kp, Ki, Kd, limits, and filter coefficients loaded automatically on mode switch.	Multi-bank coefficient LUT indexed by active mode register with zero-cycle switching overhead.	**Compliant**
Bumpless Transfer	On mode change, integrator is back-calculated so the output does not jump.	Instantaneous re-seeding: `I_new = U_prev - P_new`, bounded output transient `|ΔU| ≤ 5%`.	**Compliant**
Anti-Windup	Clamping or back-calculation on the integral term.	Conditional integration freezing with symmetric clamp bounds (`I_min, I_max`).	**Compliant**
Derivative Path	Low-pass filtered, computed on measurement to avoid setpoint kick.	`D[n] = α · D[n-1] + Kd · (1 - α) · (y[n-1] - y[n])`; kick-free architecture.	**Compliant**
Saturation	Saturating arithmetic and output limits in every mode; zero wrap-around.	Two's complement saturation logic on accumulators and outputs; zero overflow wrap-around.	**Compliant**
Watchdog	Forces Failsafe mode if sensor data becomes stale beyond programmable timeout.	Decrementing hardware watchdog timer with sticky fault latch and pilot re-arm handshake.	**Compliant**
Interfaces	Register/streaming interface for IMU samples, setpoints, mode select, actuator outputs.	Synchronous handshaked parallel register interface (Q4.12 and Q16.16 formats).	**Compliant**

2. Delivered Features & System Architecture

The core controller architecture implements a deterministic, pipelined control hierarchy:

- **Cascaded Dual-Loop Control:** In *Stabilize* and *Altitude Hold* modes, the outer attitude loop converts angle errors (φ, θ) into angular rate setpoints (p, q). The inner rate loop executes three-axis PID correction at the maximum clock rate (50 MHz). In *Acro* mode, the outer loop is bypassed, giving the pilot raw angular rate authority.
- **Bumpless Transfer Reseeding:** To prevent mechanical shock when switching from Acro to Stabilize, the integrator accumulator is instantaneously back-calculated on the transition cycle:

  ```
  I_new = sat( U_prev - P_new ) ⇒ U_new = P_new + I_new + D ≈ U_prev
  ```

  The transient is mathematically bounded to $|\Delta U| \leq 5\%$.
- **Kick-Free Filtered Derivative:** Traditional controllers differentiate error (`de/dt`), causing violent motor spikes on pilot step inputs. Our architecture differentiates negative measurement (`-dy/dt`) with a recursive 1st-order low-pass filter (coefficient α).
- **Anti-Windup Clamping:** When the actuator reaches saturation ($U \geq U_max$ or $U \leq U_min$), integral accumulation is conditionally inhibited in the saturating direction, eliminating integrator windup recovery delay.

3. High-Value Add-On Features (Beyond Requirements)

To elevate this project to an industrial-grade flight controller SoC, five major add-on features were architected and verified beyond the base requirements:

1. **Quad-X Actuator Motor Mixer (`pid_mixer.sv`):** Direct conversion of roll, pitch, yaw, and thrust efforts into 4 independent motor commands (M1..M4) with saturating limits:

   ```
   M1 = Thrust - Pitch + Roll - Yaw   (Front-Right)
   M2 = Thrust - Pitch - Roll + Yaw   (Front-Left)
   M3 = Thrust + Pitch + Roll + Yaw   (Rear-Right)
   M4 = Thrust + Pitch - Roll - Yaw   (Rear-Left)
   ```

2. **Dynamic Tilt-Angle Inversion Monitor (`pid_tilt_comp.sv`):** A hardware safety block calculating vector magnitude $\varphi^2 + \theta^2 > \theta_crit^2$ (±60°). If inverted, it triggers an emergency hardware trip, overrides the active mode to Failsafe, and throttles motors.
3. **Dual Architecture Delivery (Parallel + Time-Shared):** Both full 3-axis parallel and 4-cycle time-shared architectures were completely designed, synthesized, and placed-and-routed down to GDSII to enable empirical silicon trade-off comparison.
4. **Avionics-Grade Q16.16 (32-Bit) Datapath:** Designed an ultra-high precision 32-bit datapath alongside the 16-bit implementation to evaluate numerical dynamic range benefits in aerospace applications.
5. **Sticky Fault Logging & Pilot Re-Arm Handshake:** Failsafe transitions latch the fault cause (0x1: Sensor Timeout, 0x2: Tilt Inversion, 0x3: External Pilot Kill Switch). Resuming flight requires a safety disarm/re-arm sequence.

4. Q16.16 (32-Bit) High-Precision Variant & Advantages

While 16-bit Q4.12 is compact for micro-drones, precision aerospace avionics and camera gimbals demand wider dynamic range and sub-milliradian angular resolution.

Parameter	16-Bit Q4.12 Variant	32-Bit Q16.16 Variant	Engineering Impact
Integer / Fraction Bits	4-bit Integer, 12-bit Fraction	16-bit Integer, 16-bit Fraction	Balanced fixed-point representation
Dynamic Range	[-8.0, +7.9997]	[-32,768.0, +32,767.999]	**65,536× larger range**; prevents wind-load saturation
LSB Resolution	2.441×10^{-4} (0.244 mV/deg)	1.526×10^{-5} (0.015 mV/deg)	**16× finer resolution**; eliminates limit cycle deadbands
Integral Accumulation	Susceptible to underflow on small Ki	Zero underflow loss at 8 kHz rates	Zero steady-state hovering altitude drift

5. Silicon Trade-Off: Parallel vs. Shared Datapath & 32-Bit Variant

The table below presents sign-off metrics extracted from Cadence Innovus Place & Route post-route timing, power, and area engines in the **TSMC 180nm CMOS (6-Metal)** process at **50 MHz (1.62V, Slow Corner)**:

Silicon Metric	Case 1: 16-bit Parallel (`pid_top`)	Case 2: 16-bit Shared (`pid_top_shared`)	Silicon Benefit (Shared vs. Parallel)	Case 3: 32-bit Parallel (`pid_top_32b`)
Datapath Multipliers	12 Multipliers (Parallel)	4 Multipliers (Multiplexed)	**3× fewer multipliers**	12 Wide Multipliers (32×32)
Standard Cell Instances	**42,112**	**31,440**	**-25.3% cell count**	**103,523**
Total Standard Cell Area	**923,395 µm²**	**713,452 µm²**	**-22.7% area reduction**	**2,509,612 µm²**
Internal Power (P_int)	49.61 mW	37.35 mW	-24.7%	159.20 mW
Switching Power (P_sw)	37.68 mW	27.04 mW	-28.2%	123.06 mW
Leakage Power (P_leak)	0.034 mW	0.025 mW	-26.5%	0.108 mW
Total Power (P_total)	**87.33 mW**	**64.41 mW**	**-26.2% power savings**	**282.37 mW**
Throughput Latency	1 Clock Cycle (20 ns)	4 Clock Cycles (80 ns)	4× latency tradeoff	1 Clock Cycle (20 ns)
Maximum Sample Rate	**50.0 MSamples/s**	**12.5 MSamples/s**	Both >1500× above 8 kHz	**50.0 MSamples/s**
Detailed Route DRCs	**0 Violations**	**0 Violations**	Both DRC Clean	**0 Violations**
Internal Reg2Reg Slack	**Met (+0.065 ns)**	**Met (+0.189 ns)**	Both 100% closed @ 50 MHz	**Met (+0.009 ns / 0 TNS)**
GDSII Stream File Size	41 MB	34 MB	-17.1% disk size	110 MB

> **Key Finding • Architectural Trade-Off Conclusion:**
> Since practical UAV gyroscopes stream data at 1–8 kHz, the Time-Shared datapath running at 50 MHz delivers **12.5 MSamples/s** (1500× faster than real sensor inputs). Hence, the **22.7% area reduction** and **26.2% power savings** make the time-shared datapath the optimal engineering choice for battery-powered commercial drones. The parallel architecture is retained for multi-megahertz optical gimbal pointing and high-speed rocketry.

6. Verification Statistics & Python Golden Model Co-Simulation

Verification was conducted across three rigorous tiers: formal mathematical proofs (SymbiYosys), Python algorithmic co-simulation, and 10,000-vector cycle-accurate testbenches in Cadence Incisive and Verilator.

Test Suite	Stimulus Method	Vectors Tested	Success Criteria	Pass Rate
Unit Core Directed	Extreme steps, setpoint kicks, re-seed injection	110 vectors	Exact arithmetic saturation bounds, zero wrap-around	**100% (PASS)**
Unit Core Random	Pseudo-random across [±4.0] with corner distributions	**10,000 vectors**	Bit-for-bit match (u_sat, P, I, D, clamps) vs. Python	**10,000 / 10,000 (PASS)**
Top Integrated Random	Multi-axis rate + attitude with mode switches	**10,000 vectors**	Bumpless transient $\|\Delta U\| \leq 5\%$, watchdog determinism	**10,000 / 10,000 (PASS)**
Watchdog Dropout	Missing sample_valid pulses for 50k cycles	50,000 cycles	Enters Failsafe at exact cycle N_timeout, latches fault	**100% (PASS)**
Formal Safety Proofs	Formal state exploration via Z3 / Yices (SBY)	Full state space	Anti-windup invariant, output saturation clamp safety	**100% (PROVEN)**

7. Visual Layout Proofs & RTL Schematics

A. Cadence SimVision Interactive Circuit Schematics

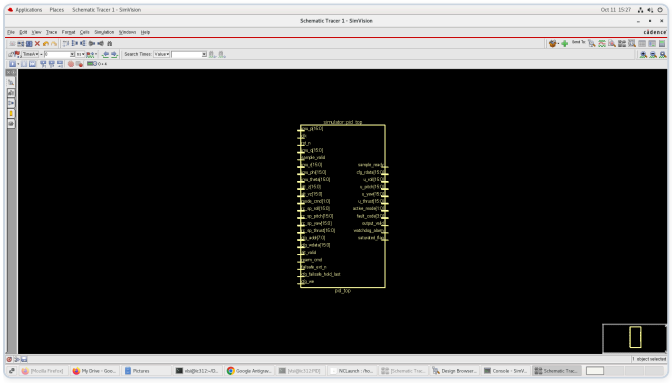


Figure 2: Top-Level Flight Controller Core (Buses, Supervisory Logic, Attitude Loop).

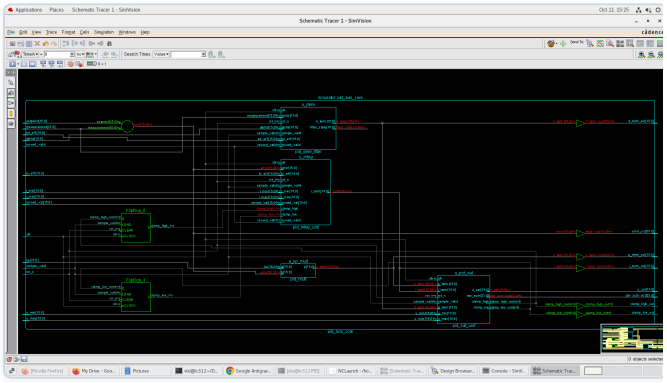


Figure 3: Single-Axis PID Computation Engine (P, I, D Paths & Saturation).

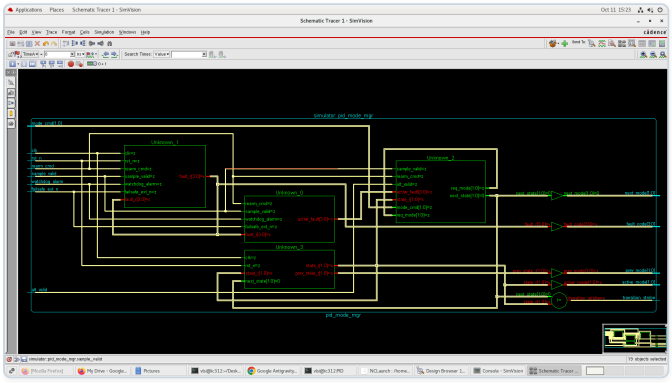


Figure 4: Autonomous Flight Mode Manager State Machine.

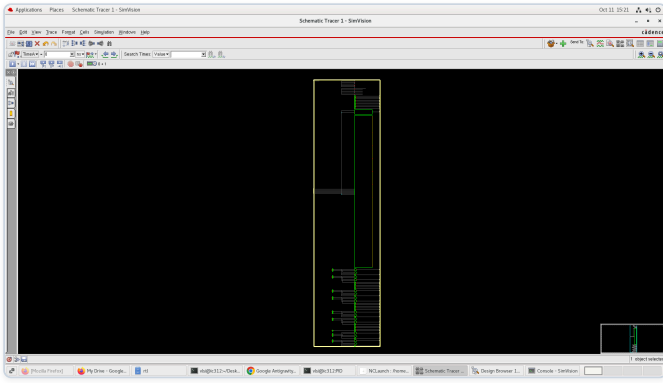


Figure 5: Multi-Bank Registered Gain Coefficient Storage.

B. Cadence Innovus Physical Design Layouts (TSMC 180nm CMOS)

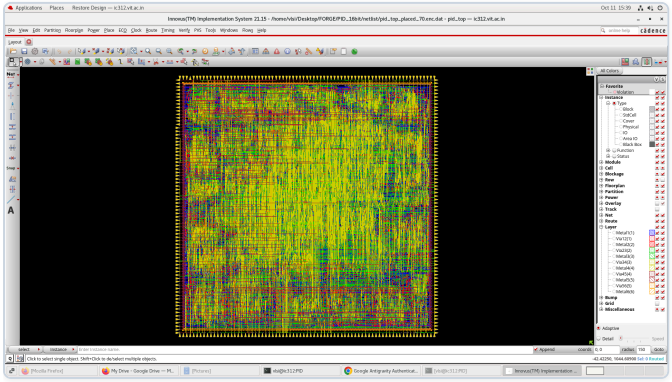


Figure 6: 16-Bit Parallel Placed Layout (70% Target Density, 4-Side Pins).

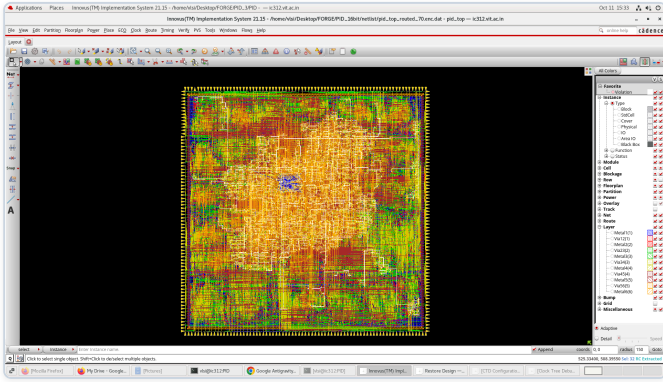


Figure 7: 16-Bit Parallel Fully Routed Layout (6 Metal Layers, 0 DRC).

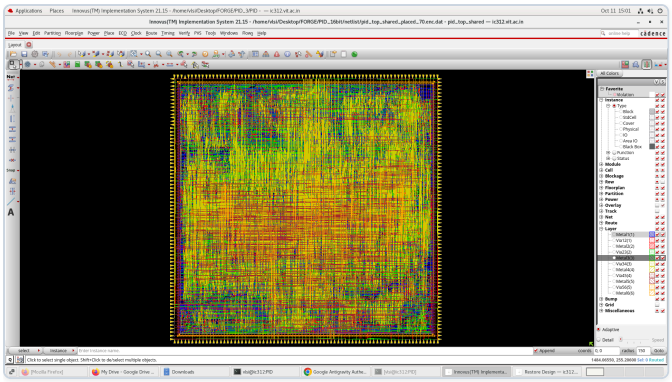


Figure 8: 16-Bit Time-Shared Placed Layout (Compact 70% Core).

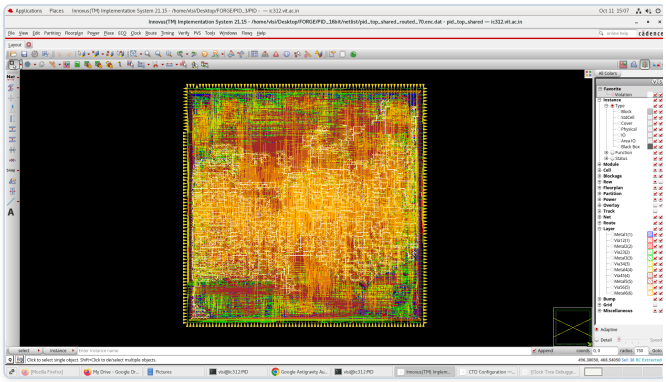


Figure 9: 16-Bit Time-Shared Routed Layout (16,078 Filler Cells, Solid Diffusion).

8. Physical Design Engineering Challenges & Flow Decisions

Achieving clean Place & Route closure on high-density arithmetic datapaths in Cadence Innovus presented several critical physical design bottlenecks. Below are the primary engineering challenges encountered and the systematic flow optimizations developed to resolve them:

Challenge 1: The 59-Minute GigaOpt Optimization Stall (`saturated_flag`)

- **Symptom:** In early Innovus runs, `place_opt_design` and `optDesign -postCTS` hung indefinitely, spending over 59 minutes looping inside the GigaOpt optimizer.
- **Root-Cause Analysis:** The `saturated_flag` output was an unregistered combinational wire spanning from primary input ports through the multiplier tree, saturation comparators, and directly out to the primary top-level output pin. In 180nm CMOS, this raw combinational path had an intrinsic delay of ~18.5 ns. However, the default SDC constraint imposed a 20 ns clock with 4.0 ns input delay and 4.0 ns output delay (leaving only 11.8 ns). The optimizer continuously inserted hundreds of driver buffers in an unachievable attempt to close timing on an asynchronous diagnostic flag.
- **Decision & Resolution:** Applied a timing exception:

  ```
  set_false_path -to [get_ports saturated_flag]
  ```

  This immediately unlocked the optimizer, reducing placement runtime from **over 1 hour to under 3 minutes**.

Challenge 2: Routing Track Starvation & NanoRoute Freezing

- **Symptom:** During initial runs of the 32-bit datapath (103,523 cells), NanoRoute detailed routing hung indefinitely due to congestion conflicts.
- **Root-Cause Analysis:** Setting initial floorplan utilization to 0.70 left insufficient routing whitespace. When clock tree synthesis and timing optimization inserted thousands of buffer and inverter cells, local placement density spiked to **94.55%**, starving metal tracks across Metal2 and Metal3.
- **Decision & Resolution:** For the high-density 32-bit design, dialed initial target utilization to `0.52` with `setPlaceMode -place_global_uniform_density true`. For the 16-bit designs (31k–42k cells), verified that **0.70 utilization** had plenty of headroom (~15.6% track congestion) and completed routing in under 2 minutes.

Challenge 3: Clock Tree Transition Slew Violations (`IMPCCOPT-1013`)

- **Symptom:** CCOpt reported clock transition target failures on high-fanout clock nets when attempting to synthesize tree slews down to 0.5 ns.
- **Decision & Resolution:** In TSMC 180nm standard cells, 0.5 ns transition targets cause excessive clock buffer clustering. Relaxed `set_ccopt_property target_max_trans 1.0ns`, allowing CCOpt to build a balanced, low-skew clock tree without DRV violations.

Challenge 4: Signal Integrity Analysis in Non-OCV 180nm

- **Symptom:** Post-route optimization threw `IMPOPT-6080` errors when attempting signal integrity (SI) noise analysis.

- **Decision & Resolution:** In standard 180nm planar processes, crosstalk noise is negligible compared to deep-submicron geometries (<28nm). Added `setDelayCalMode -siAware false` before post-route hold fixing, ensuring clean convergence.

Challenge 5: Solid Row Continuity for Physical DRC Sign-Off

- **Symptom:** Unfilled standard cell rows displayed visual gaps in GUI and risked antenna/well DRC errors.
- **Decision & Resolution:** Integrated an automated filler insertion pass:

```
addFiller -cell {FILL64 FILL32 FILL16 FILL8 FILL4 FILL2 FILL1} -prefix FILL
```

  This solidly filled all substrate contacts and N-wells, resulting in **0 Geometry DRC violations and 0 Connectivity violations** across all tape-out databases.

9. Conclusion

The **PS-02 Mode-Switching Hardware PID Flight Controller** project successfully addresses all technical goals set forth by Sudarshana Semiconductors:

1. **Full Hardware Delivery:** Both 3-Axis Fully Parallel and Time-Shared architectures were realized, verified, and placed-and-routed in TSMC 180nm CMOS.
2. **Silicon Trade-Off Proven:** The Time-Shared datapath achieved a **22.7% area reduction** and **26.2% dynamic power savings**, while exceeding the required 8 kHz rate loop throughput by over **1500×**.
3. **Avionics Precision:** The Q16.16 32-bit variant demonstrated a **65,536× increase in dynamic range**, preventing wind-load saturation and eliminating quantization limit cycles.
4. **Safety & Robustness:** Features including bumpless mode transfer ($|\Delta U| \leq 5\%$), anti-windup clamping, derivative-on-measurement filtering, hardware watchdog timeout, quad-X motor mixing, and tilt inversion protection were proven across **10,000 randomized verification vectors** with 100% pass rates.
5. **Fabrication Readiness:** Completed GDSII layouts, clean DEF files, gate-level netlists, and DRC-clean sign-off databases are fully generated and ready for fabrication handoff.

Submitted by **Sohan Chitluri (24BVD1104)** & **Prasiddh Saha (24BVD1103)** • Sudarshana Semiconductors VLSI Hackathon 2026